Arjun Nanduri

Berkeley, CA | arjunnanduri@gmail.com | (407)-580-8788 | linkedin.com/in/arjunnanduri | github.com/FavouritePlayer

Education

University of California: Berkeley, BA in Computer Science — Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Research Experience

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA — Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using a custom edit-distance metric over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (∼**1M annotated words across** ∼**10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Machine Learning Engineer, Aver Technologies Inc. – Virginia — Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Pipeline mechanism, on 1000fps footage: object detection to localize the pile, YOLO segmentation and line detection to track painted depth markings, an open-source digit detector, and audio-visual synchronization to isolate each hammer strike, **converting pixel-space line displacement into inch/foot measurements** via a depth map.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.

Research Projects

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3) — github.com/FavouritePlayer/ant_sim

- Trained PPO quadruped policies in MuJoCo/Gymnasium, benchmarking flat-ground baselines against terrain-adapted and damage-robust variants across **10 matched random seeds per condition**.
- Terrain adaptation: policy trained on randomized heightfields achieves **2.1x the reward (893±130 vs. 424±106)** and 40% lower fall rate than the flat-trained baseline on unseen rough terrain.
- Leg-damage robustness: a specialist policy reaches ∼49x the flat-baseline reward on that leg but **does not transfer to other legs (100% fall) —** documented as a negative result rather than shipped as an unsupported claim.
- Built a compound policy router that switches checkpoints by damaged leg, beating any single policy across a 4-leg × 10-seed evaluation sweep; **validated via multi-seed replication and an automated regression-test suite**.

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy) — github.com/FavouritePlayer/face-morpher

- Implemented per-triangle affine solve (3-point correspondence → 2×3 matrix), inverse warping with bilinear interpolation, and piecewise Delaunay-triangle warping to produce morphs and mean faces; **all core geometry self-implemented in NumPy**, OpenCV limited to I/O and masking.

Publications

- **Scoping Review of PTSD Treatments for Natural Disaster Survivors,** *Dec 2023*, Arjun Nanduri, et al., doi: 10.52965/001c.89642
- **Triquetrum Fracture with Pisiform Dislocation,** *Mar 2022*, Arjun Nanduri, et al., doi: 10.52965/001c.32339
- **Baker's Cyst,** *Dec 2021*, Arjun Nanduri, et al., doi: 10.7759/cureus.20403

Skills

Languages: Python, SQL

Technologies: PyTorch, Transformers (BART), scikit-learn, Pandas, NLTK, spaCy, MuJoCo, Gymnasium, Stable-Baselines3, NumPy, dlib, scipy, Git